Project Report:

Project Title: Smart Maintenance AI: Explainable Multimodal AI for Predictive Maintenance and Root-Cause Analysis

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1. Project Title

Explainable Multimodal AI for Predictive Maintenance and Root-Cause Analysis in Smart Manufacturing (Smart Maintenance AI)

The project was built in Google Colab (Python, scikit-learn, XGBoost, PyTorch, SHAP, Sentence-Transformers) and is accompanied by an interactive website that lets a user enter machine readings, get a health report, and browse every graph produced in the notebook.

2. What the Project Does

Smart Maintenance AI is a decision-support system for industrial machines. Instead of relying on one data source, it combines four kinds of evidence about a machine and turns them into a single, explainable health report:

1. **Sensor readings** (temperature, rotational speed, torque, tool wear) are used to predict the probability that the machine will fail soon, and to flag unusual operating behaviour.
2. **Sensor time-series history** is used to estimate the Remaining Useful Life (RUL), meaning how many operating cycles are left before failure.
3. **Surface images** of a component are classified into six steel-defect types, and a Grad-CAM heat-map shows which region of the image drove the decision.
4. **Free-text maintenance logs** written by operators are analysed to identify the likely fault type, its severity, and key words.

These outputs are fused into one overall risk score and machine status (Healthy, Moderate Risk, High Risk or Critical), along with SHAP-based root causes and plain-language maintenance recommendations.

2.1 Problem Statement

Unexpected machine failures cause production downtime, high maintenance cost, equipment damage and safety risks. Reactive and preventive maintenance cannot predict failures accurately in advance.

Most predictive-maintenance systems also depend on a single data source (sensor readings) and say little about the cause of a failure. Industrial data actually arrives in several forms (sensor streams, maintenance reports and machine images), and combining them is hard. The project therefore proposes an explainable multimodal AI system that predicts failures, estimates remaining useful life, identifies probable root causes and recommends maintenance actions.

2.2 Objectives

1. Predict machine failures before they occur using industrial sensor data.
2. Detect abnormal machine behaviour with unsupervised anomaly detection.
3. Estimate Remaining Useful Life (RUL) with a time-series deep-learning model.
4. Analyse maintenance reports with NLP to find symptoms, components and previous maintenance information.
5. Analyse industrial images with computer vision to identify visible abnormalities.
6. Combine the modalities through multimodal fusion.

7. Identify the probable root cause of a predicted failure.
8. Apply Explainable AI (SHAP) to show why a prediction was made.
9. Generate actionable maintenance recommendations.

3. Dataset Description

Three public datasets are used, one per data modality.

Dataset	Modality	Size and content	Used for
AI4I 2020 Predictive Maintenance	Tabular sensor data	10,000 records, 14 columns. No missing values and no duplicates. 339 failures (3.39%) and 9,661 non-failures, so the data is strongly imbalanced.	Failure prediction, anomaly detection, operating-state clustering
NASA C-MAPSS (subset FD001)	Multivariate time series	20,631 rows (26 columns) of turbofan-engine run-to-failure data: 3 operational settings and 21 sensors per cycle.	Remaining Useful Life regression
NEU-DET surface defects	Images	1,800 steel-surface images in 6 balanced classes (crazing, inclusion, patches, pitted_surface, rolled-in_scale, scratches). 240 training and 60 validation images per class.	Visual defect classification

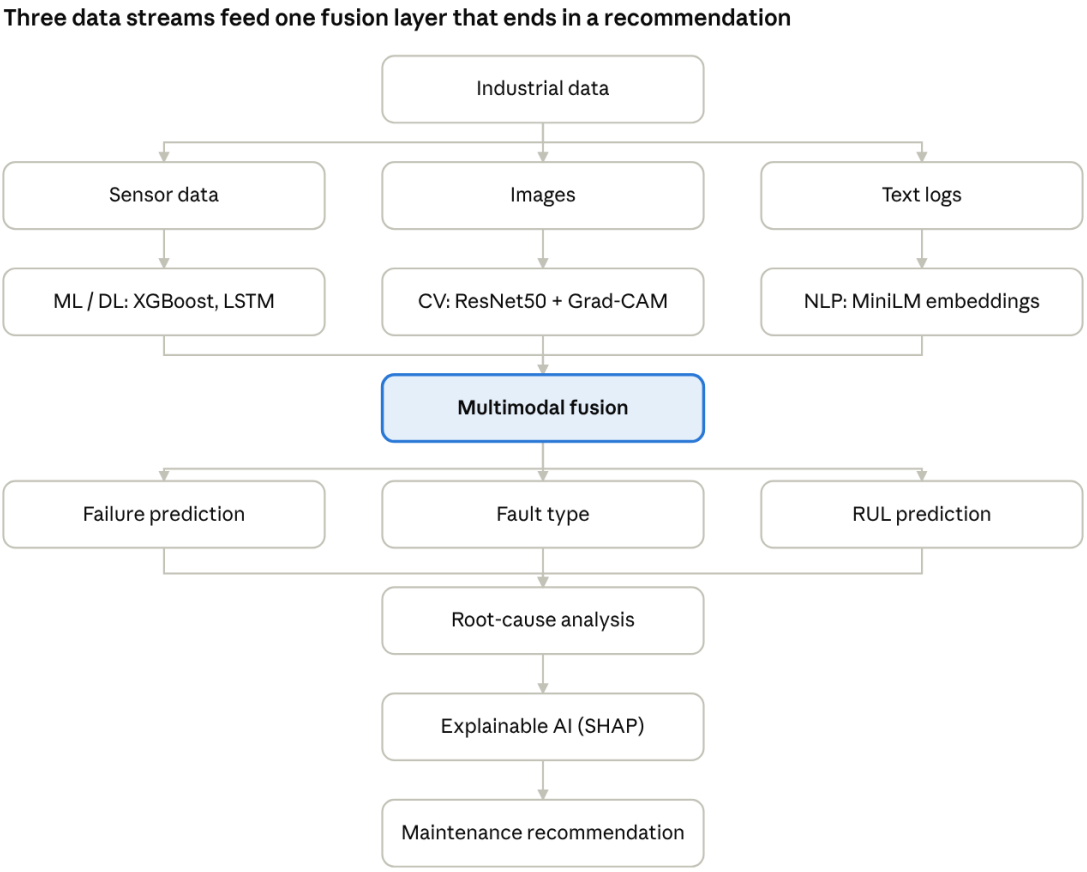
AI4I columns: Product ID and UDI (identifiers), Type (L/M/H quality variant), Air temperature [K], Process temperature [K], Rotational speed [rpm], Torque [Nm], Tool wear [min], the target Machine failure, and five failure-mode flags: TWF (tool wear), HDF (heat dissipation), PWF (power), OSF (overstrain) and RNF (random). Torque ranges from 3.8 to 76.6 Nm, and tool wear from 0 to 253 minutes.

C-MAPSS: RUL is not given for training data, so it was derived as (last cycle of the engine) minus (current cycle). Of the 21 sensors, 19 have non-zero variance and were kept.

Maintenance logs: are not drawn from a public dataset. The NLP module is zero-shot, and the project was tested with hand-written example logs.

4. Methodology

The system is organised as four independent analysis modules feeding one fusion layer.



proposed methodology · 3 inputs, fusion, 3 outputs, recommendation

The workflow follows the proposal: each data type goes through its own model, the fusion layer merges them into failure, fault-type and RUL outputs, and root-cause analysis, explainability and a maintenance recommendation follow.

4.1 Sensor module (AI4I)

- **Preprocessing:** identifier columns were dropped, *Type* was one-hot encoded, and an 80/20 stratified train/test split was made (8,000 / 2,000 samples, seed 42). Features were standardised with `StandardScalar`.
- **Models compared:** Logistic Regression, Decision Tree, Random Forest (300 trees, balanced class weights) and XGBoost (300 trees, depth 5, learning rate 0.05, subsample 0.8, column-sample 0.8).
- **Evaluation:** accuracy, precision, recall, F1, ROC-AUC, confusion matrix and ROC curves.
- **Explainability:** feature importance and **SHAP** (TreeExplainer) summary and per-machine waterfall plots.
- **Unsupervised analysis: Isolation Forest** (200 trees, 5% contamination) for anomaly detection, and **K-Means** (k = 3) for operating states, visualised with PCA.

4.2 Remaining Useful Life module (C-MAPSS)

- RUL labels were created, constant sensors were removed, and the data was scaled with a MinMax scaler.
- Sliding windows of 30 cycles × 19 sensors were built per engine, giving 17,631 sequences (14,104 train / 3,527 validation).
- A **PyTorch LSTM** (hidden size 64, followed by one linear layer) was trained with MSE loss, Adam (lr 0.001), batch size 64, for 15 epochs on a GPU.

4.3 Computer-vision module (NEU-DET)

- Images were resized to 224×224, converted to 3-channel greyscale, and augmented (random horizontal flip, 10° rotation).
- **Transfer learning:** an ImageNet-pretrained **ResNet50** with its backbone frozen and a new 6-class output layer, trained with cross-entropy and Adam (lr 0.0001) for 5 epochs.
- **Grad-CAM** was implemented on the last ResNet50 bottleneck block to highlight the defect region.

4.4 Maintenance-log NLP module

- The pretrained all-MiniLM-L6-v2 sentence-embedding model encodes the log text and ten industrial fault categories (Bearing Failure, Overheating, Excessive Vibration, Lubrication Problem, Motor Failure, Electrical Fault, Gear Failure, Pressure Problem, Leakage, Unknown Fault).
- The fault is the category with the highest cosine similarity to the log (zero-shot, no training needed).
- Severity (Low / Medium / High) and key words come from a keyword dictionary, and a fault-specific recommendation is attached.

4.5 Multimodal fusion

Each module is converted to a 0–1 risk and combined with a weighted sum:

Source	How risk is derived	Weight
Sensor	XGBoost failure probability	0.35
RUL	Banded: ≤10 cycles = 1.00, ≤25 = 0.75, ≤50 = 0.50, ≤100 = 0.25, otherwise 0	0.30
Image	Base risk per defect class (0.60–0.80) × model confidence	0.20
Maintenance log	Severity: Low 0.25, Medium 0.60, High 1.00	0.15

Status thresholds: risk ≥ 0.75 is *Critical*, ≥ 0.50 *High Risk*, ≥ 0.25 *Moderate Risk*, otherwise *Healthy*. Rules then produce a list of recommendations (for example "Schedule immediate machine inspection").

4.6 Software Modules

The presentation defines nine software modules. The table shows how each one is realised in the notebook.

Module	Purpose (from the proposal)	Implementation in the notebook	Status
Data preprocessing and EDA	Clean, analyse and prepare industrial data	Distributions, correlation matrix, missing/duplicate checks, encoding, scaling, stratified split	Done
Failure prediction	Predict failure probability and type	XGBoost, Random Forest, Decision Tree, Logistic Regression. Predicts failure vs no failure only, not the failure type	Done
Anomaly detection	Find unusual behaviour	Isolation Forest, K-Means operating states, PCA plot	Done
RUL prediction	Estimate remaining life with deep learning	LSTM on 30-cycle windows of C-MAPSS FD001	Done
NLP analysis	Extract failure information from logs	MiniLM embeddings for fault type, keyword rules for severity. Components and past maintenance history are not extracted	Done
Multimodal fusion	Combine sensor, text and image information	Weighted risk fusion (0.35 / 0.30 / 0.20 / 0.15) of sensor, RUL, image and log risks	Done
Root-cause analysis	Identify probable reasons for failure	Top SHAP factors, visible image defect and log-derived fault	Done
Explainable AI	Explain each prediction	SHAP summary and waterfall plots, Grad-CAM heat-maps	Done
Maintenance recommendation	Suggest actions	Rule-based recommendations from risk levels and fault type	Done

4.7 Challenges and How the Project Handles Them

Challenge (from the proposal)	How it is addressed
Heterogeneous data	One specialised model per modality, joined by a common 0–1 risk scale in the fusion layer
Noisy and incomplete data	Checks found no missing values or duplicates in AI4I. Constant C-MAPSS sensors were dropped and all inputs scaled
Rare failures (class imbalance)	Only 3.39% of AI4I rows are failures, so the split was stratified and Random Forest used balanced class weights

Challenge (from the proposal)	How it is addressed
Time-dependent behaviour	Sliding 30-cycle windows with an LSTM
Root-cause identification	SHAP contributions, Grad-CAM region and log fault type are reported together
Explainability	SHAP for tabular data, Grad-CAM for images, matched keywords for text
Multimodal fusion	Weighted risk score with clear status thresholds

5. Novelty

- **Four modalities in one pipeline:** Tabular sensors, time series, images and free text are rarely combined in a student predictive-maintenance project. Most work addresses only one.
- **From prediction to action:** The output is not only a probability. It includes root causes, a status, and recommended maintenance steps, so the system supports a decision.
- **Explainability throughout:** SHAP explains the sensor model, Grad-CAM explains the image model, and extracted keywords and similarity scores explain the text model.
- **Zero-shot maintenance-log understanding:** Sentence embeddings classify logs without any labelled training data.
- **Failure prediction plus remaining life plus anomaly detection:** are provided together, covering "will it fail", "when will it fail" and "is it behaving abnormally".
- **Deployable demo:** A web interface with a fusion calculator and a graph gallery makes the work usable by non-programmers.

6. Implementation

Environment: Google Colab with Google Drive storage. A structured project folder (datasets, models, preprocessing, results, checkpoints, reports) was created and models, scalers, figures, tables and JSON reports were saved at each stage so that later steps could reload them.

Step-by-step flow

1. **Setup:** install xgboost, shap, joblib, seaborn and PyTorch, mount Drive, and create folders.
2. **AI4I:** load the CSV, run exploratory analysis (distributions, correlation matrix, failures by product type), preprocess, train the four models, compare, and save the best (XGBoost).
3. **Explainability and unsupervised learning:** SHAP plots, Isolation Forest anomaly flags (98 of 2,000 test samples flagged) and K-Means states (clusters of 5,550 / 2,364 / 86 training samples).
4. **Prediction function:** a reusable routine takes a machine's readings, aligns them to the 12 features XGBoost expects, and returns failure probability, risk level, anomaly status, top-5 SHAP root causes and a recommendation. It outputs a formatted *Smart Machine Health Report* and a JSON file.

5. **C-MAPSS:** build RUL labels, windows and the LSTM, then evaluate with MAE and RMSE.
6. **Image classifier:** extract NEU-DET, build data loaders, fine-tune the ResNet50 classifier head, evaluate, and add Grad-CAM with a reusable `analyse_image` function.
7. **NLP:** encode the fault categories and write functions for fault, severity, keywords and recommendation.
8. **Fusion:** `integrated_machine_analysis()` combines the four risks and returns the status, overall risk and recommendations, and a final human-readable report and CSV summary are saved.
9. **Website:** the notebook's results were packaged into an interactive page. The user enters readings (product type, sensor failure probability, RUL, image defect and confidence, maintenance log) and gets the combined assessment. Graphs from the notebook can be opened full size. Values marked *Notebook output* come directly from the notebook, and the rest are in-browser estimates.

7. Results and Discussion

7.1 Failure prediction (AI4I test set, 2,000 samples, 68 failures)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9990	1.000	0.971	0.985	0.973
Decision Tree	0.9960	0.917	0.971	0.943	0.985
Random Forest	0.9985	1.000	0.956	0.977	0.991
XGBoost	**0.9990**	**1.000**	**0.971**	**0.985**	**0.996**

XGBoost was selected for the highest ROC-AUC. Its confusion matrix shows 1,932 true negatives, 0 false positives, 66 true positives and 2 missed failures. Feature importance was dominated by the failure-mode flags (HDF 0.31, PWF 0.23, OSF 0.21, TWF 0.16), followed by torque, rotational speed and tool wear.

7.2 Remaining Useful Life (C-MAPSS FD001)

The LSTM's validation loss fell steadily from 9,174 to 1,507 (MSE) over 15 epochs, and final validation errors were **MAE = 25.07 cycles** and **RMSE = 38.81 cycles**. The model learns the degradation trend, but an average error of about 25 cycles is large, so the RUL estimate is better suited to coarse risk banding (as used in the fusion step) than to precise scheduling.

7.3 Image defect classification (NEU-DET validation, 360 images)

Accuracy 91.4%, precision 0.916, recall 0.914, F1 0.911. Accuracy rose from 62.8% to 91.7% within four epochs and then levelled off.

Class	F1	Comment
rolled-in_scale	1.00	Perfectly separated

Class	F1	Comment
patches	0.95	Strong
scratches	0.94	Recall 1.00, but some other classes confused with it
crazing	0.93	Strong
pitted_surface	0.84	Moderate
inclusion	0.80	Weakest (recall 0.70)

Per-image softmax confidences are low (for example 32–43% on some correct predictions) because only the final layer was trained for a short time. The predicted class is usually right, but the confidence value should not be read as a calibrated probability.

7.4 NLP maintenance-log analysis

On hand-written test logs, the module identified the expected fault: "excessive vibration and unusual noise" gave *Excessive Vibration* (74.8%), "motor temperature very high, cooling fan stopped" gave *Overheating* (41.6%, High severity), and "oil leaking from hydraulic pipeline" gave *Leakage* (71.6%, High). Confidences ranged from about 41% to 75%. This is a qualitative check only, because there is no labelled log dataset to compute accuracy.

7.5 Anomaly detection and operating states

Isolation Forest flagged 98 of 2,000 test machines (about 4.9%), consistent with its 5% contamination setting. K-Means separated one dominant normal state, one mid-sized state and a small cluster of 86 machines that is the most likely group of unusual operating conditions.

7.6 Multimodal fusion demonstrations

Scenario	Sensor	RUL	Image	Log severity	Overall risk	Status
Severe case	72.5%	18	scratches, 96.8%	High (bearing failure)	75.46%	Critical
Real image + log	72.5%	18	pitted surface, 32.4%	High	68.05%	High Risk
Healthy machine (early fusion demo)	0.48%	30%	20%	n/a	15.19%	Normal

The fusion shows the intended behaviour: when several independent sources agree on a problem, the status escalates, and a healthy sensor reading with low other risks stays Normal. The reports also list the specific recommendations triggered by each source.

8. Conclusion

Smart Maintenance AI shows that sensor data, degradation time series, component images and operator notes can be combined into one explainable view of machine health. XGBoost gave the best failure classifier (ROC-AUC 0.996), the LSTM learned remaining-life trends on NASA turbofan data (MAE 25.1 cycles), the ResNet50 detector reached 91.4% accuracy on six steel-defect types, and a zero-shot sentence-embedding model turned free-text logs into fault types and severities. SHAP and Grad-CAM make the individual predictions inspectable, and the fusion layer turns everything into a status and actionable recommendations, delivered through an interactive website.

Measured against the nine objectives in Section 2.2, seven are delivered (failure prediction, anomaly detection, RUL estimation, image analysis, fusion, SHAP explanations and recommendations). Two are delivered in simplified form: NLP extracts fault type, severity and keywords but not components or maintenance history, and root-cause analysis points to the strongest contributing factors rather than a verified cause.

The project is a strong working prototype. The most valuable next steps are to remove the failure-mode flags from the AI4I features, fix the probability-unit handling in the fusion step, evaluate the LSTM on engine-wise and official test splits, fine-tune the image model further, and learn or calibrate the fusion weights using data in which all four modalities come from the same machine. With these changes the reported performance would better reflect real-world use.

References (datasets and tools)

- AI4I 2020 Predictive Maintenance Dataset (UCI Machine Learning Repository)
- NASA C-MAPSS Turbofan Engine Degradation Simulation Dataset (NASA Prognostics Data Repository)
- NEU-DET Steel Surface Defect Database (Northeastern University)
- XGBoost, SHAP, scikit-learn, PyTorch / torchvision (ResNet50), Sentence-Transformers (all-MiniLM-L6-v2)

Reference papers (from the project presentation)

- A Systematic Literature Review of Machine Learning Methods Applied to Predictive Maintenance (2019)
- Smart Fault Diagnosis with Deep Learning for Predictive Maintenance
- Interpretable Fault Detection Approach With Deep Neural Networks to Industrial Applications
- Deep, Generative, and Explainable Models for Anomaly Detection in Smart Manufacturing
- Explainable Predictive Maintenance of Rotating Machines Using LIME, SHAP, PDP, ICE, IEEE Access, vol. 12, pp. 29345–29361, 2024